International Islamic University Chittagong

Department of Computer Science and Engineering
Final Term Examination, Spring-2024
Course Title: Compiler
Course Code: CSE-3527

Full Marks: 50 Time: 2 Hours 30 Minutes

Answer all Five of the following questions.
Figures in the right hand margin indicate full marks.

Group A

1. a) Write the role of a parser. Draw a hierarchy diagram to show all types of common parsers used in compiler construction. 2 CO2

b) What is the rule of left recursion in compiler? Is there any left recursion in the following grammar G_1? If yes, Eliminate the left recursion. 3 CO3

$$S \rightarrow S - A | A$$
$$B \rightarrow B + C | C$$
$$C \rightarrow id$$

c) Consider the following CFG G_2 over the language Σ = {int, (,), +, *}. 5 CO2

$$E \rightarrow TX$$
$$T \rightarrow int\ Y |\ (E)$$
$$X \rightarrow +E | \epsilon$$
$$Y \rightarrow * T | \epsilon$$

i) Find out the FIRST and FOLLOW set for each of the non-terminals in the above grammar

ii) Construct the predictive parsing table based on the above grammar G_2 *and show the stack implementation along with output parse tree for the input string* "*int*int*". Is this grammar LL (1)? Explain.

2. a) What is the basic difference of LR (0), SLR(1), CLR(1) and LALR parser tables? Why '\$' symbol uses as an element of follow set of the start variable of a grammar in the parser? 2 CO3

b) Consider the following grammar G_2 over the alphabet $\Sigma = \{1, -, *\}$. You want to use an SLR(1) or CLR(1) parser to parse strings defined by G_2. Notice that each production in G_2 is associated with a production number in first bracket. 8 CO3

$$(1)\ S \rightarrow N * N$$
$$(2)\ S \rightarrow N * P$$
$$(3)\ N \rightarrow\ -P$$
$$(4)\ P \rightarrow 1P$$
$$(5)\ P \rightarrow 1$$

I_0:

$S' \rightarrow .S$
$S \rightarrow .N * N$
$S \rightarrow .N * P$
$N \rightarrow .-P$

Fig. 2(a)

I_0:

$S' \rightarrow .S, \$$
$S \rightarrow .N * N, \$$
$S \rightarrow .N * P, \$$
$N \rightarrow .-P, *$

Fig. 2(b)

OR

a) Construct the LR (0) parsing table for the following grammar: 3 CO3

S → AA

A → aA/b

b) Parse the following string using the parsing table you just made in 2(a). 5 CO3

abab$

c) Give bottom-up parsing for the input string aa*a+ for the following grammar: 2 CO3

S → SS+ | SS* | a

Group B

3.

a) What is the basic difference between synthesized and inherited attributes for syntax-directed translation? Consider the following CFG G3 over the language Σ = {int, float, char, double, id,','}. 5 CO2

$D \rightarrow TL$

$T \rightarrow$ **int** |**float**|**char**|double

$L \rightarrow L_1$, **id** | id

Write semantic rules for syntax-directed translation of the above grammar G3. Construct the dependency graph for the input string "**double a, b, c**".

b) Write the impact of the type-checker on compiler construction? Write semantic rules for the following type checking expressions and statements: 5 CO3

$E \rightarrow E_1 mod E_2$

$E \rightarrow E_1\ op\ E_2$

$S \rightarrow if\ E\ then\ S_1$

$S \rightarrow while\ E\ S_1$

4.

a) What is the activation tree in the run-time environment of a compiler for a function? Explain the memory allocation strategies for the run-time environment. Draw the activation tree for the following c programming function if n=6: 5 CO3

```
long factorial (int n) {
  if (n==1)
    return 1;
  else
    return n*factorial(n-1);
}
```

b) Translate the arithmetic expression $o := x * (-y) + x * (-y)+x*y$ into 5 CO3

i) Syntax tree

ii) Three-address code

iii) Triples

iv) DAG

v) Quadruples

5.

a) Consider the following C codes and answer the questions (i) to (iii): 5 CO3

```
void insertionSort(int array[], int n){
   int i, key, j;
   for (i = 1; i < n; i++){
      key = array[i];
      j = i - 1;
      while (j >= 0 && array[j] >key){
         array[j+1] = array[j];
         j=j-1;
      }
      array[j+1] = key;
   }
}
```

i) Translate the C code into three-address code.

ii) Identify the basic block in three-address code.

iii) Construct the flow graph from the three-address code.

b) Write some code optimization techniques used in compiler construction with examples and Construct the dag for the following basic block. 5 CO3

$$d := b * c$$
$$e := a + b$$
$$b := b * c$$
$$a := e - d$$

OR

a) Consider the following C program (Answer any one): 7 CO2

(i)for I from 1 to 10 do
for j from 1 to 10 do
a[i, *j*] = 0.0;
for *I* from 1 to 10 do
a[i, j] = 1.0;

```
(ii) for (i=0; i<n; i++)
        for (j=0; j<n; j++)
           c[i][j] = 0.0;
     for (i=0; i<n; i++)
        for (j=0; j<n; j++)
           for (k=0; k<n; k++)
              c[i][j] = c[i][j] + a[i][k]*b[k][j];
```

Translate the above C program into three address code. Also construct basic block and flow graph for the above C code.

b) Draw the syntax tree and DAG for the following expression: 3 CO2

a+a+(a+a+a+(a+a+a+a))

International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering (CSE)

B.Sc in CSE Final Examination, **Spring - 2024**

Course Code: EEE-2421 **Course Title:** Electrical Drives and Instrumentation

Full Marks: 50 **Time:** 2 hour 30 minutes

[Answer all the five questions. Figures in the right-hand margin indicate full marks]

Course outcomes and Bloom's Levels are mentioned in additional columns

	Part-A	**CLO**	**DL**	**Marks**
1(a)	"Sometimes Induction Motor is called as Rotating Transformer."-Justify the statement. Briefly Explain the working principle of the induction motor. Also draw a neat diagram of its Exact equivalent circuit.	CO2	U	1+4
1(b)	A 480-V, 50-Hz, 50-hp, three-phase induction motor is drawing 60 A at 0.85 PF lagging. The stator copper losses are 2.1 kW, and the rotor copper losses are 300 W. The friction and wind age losses are 500 W, the core losses are 1800 W, and the stray losses are negligible. Find the following quantities: **i.** The air-gap power **ii.** The power converted **iii.** The output power **iv.** The efficiency of the motor	CO3	An	5
	OR			
1(b)	A 208-V, 10-hp, four-pole, 60-Hz, Y- connected induction motor has a full-load slip of 5 percent i) What is the synchronous speed of this motor? ii) What is the rotor speed of this motor at the rated load? *iii) What is the rotor frequency of this motor at the rated load?* *iv) What is the shaft torque of this motor at the rated load?*	CO3	An	5
2(a)	*What do you understand by the term "Slip"? A 6-pole, 3-phase induction motor operates from a supply whose frequency is 60 Hz. Calculate:* *i.* Magnetic field speed **ii.** Rotor speed **iii.** Frequency of rotor current at standstill.	CO3	An	3
	OR			
2(a)	A hybrid VR stepping motor has 8 main poles which have been castellated to have 5 teeth each. If the rotor has 50 teeth, calculate- **i.** Step angle **ii.** Resolution	CO3	An	3
2(b)	How to make a synchronous motor self-starting? Explain.	CO2	U	2
	OR			
2(b)	Write different types of Induction motor rotor and describe reasons behind this type of naming.	CO2	U	2
2(c)	When a $3-\varphi$ power supply is connected with the stator of a $3\text{-}\varphi$ Induction Motor then a rotating magnetic field creates and rotates around the stator. Explain how it happens? Also draw the transformer equivalent circuit of induction motor.	CO2	U	5

Part-B

3(a)	What do you understand by the term measurement and instrumentation? Why measurement is so important in our life?	CO2	U	3
	OR			
3(a)	What is Measurement? Briefly explain the steps of generalized measuring system with proper example.	CO2	U	3
3(b)	Why Measurement accuracy and precision need in a measurement system?	CO2	U	2
3(c)	Suppose you have a voltage generator. You want to measure the frequency of the voltage signal using digital technique. But you do not have any device to measure frequency. So, you decided to design a frequency counter. Draw the block diagram of circuit and explain each block.	CO3	An	5
4(a)	The accuracy of five precision resistors is checked by comparing each of them to a 1.0 Ω standard resistor. The measured resistances are as follows: R1=1.001 Ω, R2=1.002 Ω, R3=0.999 Ω, R4=0.998 Ω and R5=1.000 Ω. Calculate: **i.** The average measured resistance **ii.** The average deviation **iii.** The standard deviation **iv.** The probable measurement error	CO3	An	5
4(b)	Explain Photo-Voltaic effect briefly.	CO2	U	~~2~~ 3
	OR			
4(b)	Briefly explain the working principle of PV cell. Also mention advantages and disadvantages of PV cell.	CO2	U	~~2~~ 3
4(c)	An op-amp has a common mode gain 5. If 20v and 16v signals are given to input terminal, then a 40v output has come. Write your comments on whether this is a good op-amp or not good op-amp based on your analysis.	CO3	An	5 2
5(a)	Briefly explain the working of Resistance Temperature Detector (RTD) and also write the applications of RTD.	CO4	U	5
	OR			
5(a)	What is transducer? Explain the working principle of Piezoelectric sensor with appropriate figure.	CO4	U	5
5(b)	Consider two voltage measurements V1 and V2 with the following specified values and tolerances: • $V1=100V\pm1\%$ • $V2=80V\pm5\%$ **i.** Calculate the maximum percentage error of two voltage measurements **ii.** Calculate the maximum percentage error in the difference of two voltage measured voltages	CO3	An	5

International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in CSE Final Examination, Spring-2024
Course Code: CSE-3529 Course Title: System Analysis and Design
Marks: 50 Time: 2 Hours 30 Minutes

Instructions: *Answer all the questions. Figures on the right margin indicate marks.*

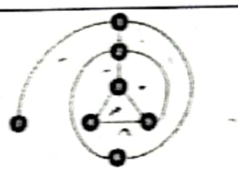
GROUP: A

1	a)	Suppose, "MetroTrain" is one of the most popular metro train systems in the world. This system bears the following characteristics: i. Users can register for a personal account with the necessary personal information, username, and password to access the metro train services. ii. Users can purchase tickets, check train schedules, and view other related information from their personal profiles. iii. All users have to log in to the system going through an authentication process. Based on the above scenario, let's identify: A. Find all the external entities B. Design Context diagram or Level-0 Diagram C. Design Level-1 diagram using balancing and leveling techniques	5	CLO3
	b)	An IT company uses a project management system to assign projects to teams based on several criteria. The system evaluates project requirements and team capabilities to make decisions. The criteria used for decision-making include: • Project Complexity (Complex, Moderate, Simple) • Team Expertise (High, Medium, Low) • Deadline Urgency (Urgent, Standard, Flexible) • Budget Availability (High, Medium, Low) Based on these criteria, the system assigns the project to one of three teams: **Team A:** Suitable for complex projects with high expertise and urgent deadlines. **Team B:** Suitable for moderate projects with medium expertise and standard deadlines. **Team C:** Suitable for simple projects with low expertise and flexible deadlines. Your task is to draw a decision tree and a decision table based on this scenario.	5	CLO3

2	a)	Distinguish between bottom-up and top-down system design methodologies with proper examples and figures.	5	CO2
	b)	An airline reservation system is designed to manage flights, passengers, bookings, and other related information. The system handles various aspects of airline operations, including flight schedules, passenger details, and seat reservations. The key elements of the system include: **Airlines:** Each airline operates multiple flights. **Flights:** Each flight has a unique flight number, departure and arrival airports, and a scheduled departure and arrival time. **Passengers:** Passengers book flights and each passenger has a unique passenger ID, name, contact information, and passport number. **Bookings:** Bookings are made by passengers for flights. Each booking has a unique booking ID, booking date, and status (confirmed, canceled, pending). **Seats:** Each flight has a specific number of seats, each with a seat number and class (economy, business, first class). Based on this scenario, you need to draw an Entity-Relationship Diagram (ERD) with the proper symbol.	5	CO4
		OR		
	a)	There are three types of accounts in a bank. Assume that the accounts are current account, saving account and fixed deposit account. In current account bank do not give any interest, in saving account bank gives 5% of interest, and in fixed deposit account bank gives interest depend on time at following: **Duration** **Interest** **6 months** **8%** **1 year** **10%** **5 years** **12%** Draw the decision table, decision tree.	5	CLO3
	b)	A company wants to develop a ride-hailing system similar to Uber. The system needs to manage drivers, customers, rides, and vehicles. The following details are provided for the system: Drivers: Each driver has a unique driver ID, name, email, phone number, and license number. Customers: Each customer has a unique customer ID, name, email, and phone number. Vehicles: Each vehicle has a unique vehicle ID, license plate number, make, model, and year. Rides: Each ride has a unique ride ID, a pickup location, a drop-off location, a fare amount, and a timestamp for the start and end of the ride.	5	CLO3

		Each ride is associated with one driver, one customer, and one vehicle. Based on the provided scenario, develop an ERD that captures the relationships between drivers, customers, rides, and vehicles. The diagram should include entities, attributes, primary keys, foreign keys, and relationships with appropriate cardinalities.		

GROUP: B

3	a)	 a) Considering the above graph, answer the following question; i. Define flow graph. ii. Identifying the node, edge and region from the above flow graph. iii. Calculate the cyclomatic complexity from the above flow graph at least two ways and also show that both ways generate the same result.	5	CLO2
	b)	b) What is white box testing and black box testing? Describe with an example.	5	CLO3

4	a)	a) What is system maintenance? Demonstrate the four types of system maintenance with necessary examples.	5	CLO2
	b)	b) An EdTech company has developed a Learning Management System (LMS) to facilitate online education for schools and universities. The system includes functionalities for course management, user management (students, teachers, and administrators), content delivery, assessments, and reporting. Over time, as the system is used, it requires maintenance to fix bugs, add new features, and improve performance. write the system maintenance steps for the above scenario	5	CLO3
		OR		
	a)	Explain the process of software system implementation in the software development life cycle. What are the key activities involved in the implementation phase?	3	CLO2
	b)	What are the different types of software maintenance activities? Provide examples of corrective, adaptive, perfective, and preventive maintenance.	3	CLO2
	c)	Discuss the significance of testing in software system implementation. What types of testing should be conducted during the implementation phase, and how do they contribute to the overall quality of the software system?	4	CLO2

5	a)	Write some criteria for software selection and hardware selection	2	CLO1
	b)	Outline the difference between a security vulnerability and a security threat. Provide examples of common security vulnerabilities and threats faced by organizations today.	4	CLO4
	c)	Write short notes on: (Any two) i) Alpha Beta testing ii) Interruption, Interception iii) Computer Criminals iv) Security goals	4	CLO1

International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Spring 2024
Course Code: CSE 3521. Course Title: Computer Architecture
Total marks: **50** Time: **2.5 hours**

Answer all **Five** of the following questions. The figures in the right-hand margin indicate full marks.

Group - A

		M	C	D
1.a)	Describe the edge triggered clocking methodology.	4	1	
b)	Suppose you are given an instruction add \$s1,\$t3,\$t4. Describe the complete operation of the datapath using the instruction with a figure.	4	1	
c)	Differentiate multicycle over single cycle datapath?	2	2	
2.a)	Points out different steps and benefits of fetch execute cycle	3	5	
b)	Design and describe the working procedure of jump instruction in single-cycle data path.	7	5	
	OR			
2.a)	Write down the basic characteristics of RISc and CISC architectures .	4	1	
b)	What is exception? How exception are handled?	4	1	
c)	What temporary registers are used in multicycle datapath?	2	2	

Group - B

		M	C	D
3 a)	Does the use of pipelining reduce, increase or leave unchanged the execution time of a single machine instruction?	3	3	
b)	How can you resolve data hazard -explain with figure.	5	1	
c)	Dynamic pipelining is more complicated than the traditional or static pipelining – why?	2	2	
4.a)	How does spatial locality be helpful in handling cache misses? Explain with example.	3	3	
b)	Explain the working principle of direct mapped cache with figure. How many total bits are needed for a direct mapped cache with 64KB data in 1 word blocks?	7	3	
5 a)	Explain the working principle of TLB with figures.	5	3	
b)	What is the difference between synchronous bus and asynchronous bus? We want to compare the maximum bandwidth for a sync and async bus. The sync bus has a clock cycle time of 50ns and each bus transmission takes 1 cycle. The async bus requires 40ns per handshake. Each bus has 32 data lines. What is the bandwidth when performing one word reads from a 200ns memory?	5	4	
	OR			
5 a)	Explain cache read hit/miss.	2	1	
b)	Define miss penalty, cache miss, hit rate, tag in caching.	4	1	
c)	What is DMA controller? Describe the configuration of DMA controller.	4	1	

International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc in CSE — Final Examination, Spring-2024
Course Code: CSE-3523
Course Title: Microprocessors, Microcontrollers & Embedded Systems
Time: 2 hours 30 minutes — **Total Marks:** 50

Answer all Five of the following questions. The figures in the right-hand margin indicate full marks.

Group - A

		M	C	P	D
1.a)	Describe how you would configure 74ALS244 to apply the interrupt vector type number B7H to the data bus in response to an interrupt request (INTR). Include the necessary connections and describe it.	4	3	2	A
b)	What are the practical advantages and challenges of expanding the number of interrupt request lines from one to seven in a Microcontrollers system? How does this impact the overall system's interrupt handling efficiency?	4	3	2	A
c)	Describe the role of the Non-Maskable Interrupt (NMI) in a microprocessor system. Why is it critical for handling urgent tasks?	2	3	2	U

OR

1. Read the passage and answer questions 1(a-b):
Suppose you have 4 input devices named Disk, Printer, Reader, and Keyboard. All devices are sending interrupts to the Microprocessor 8086.
N.B: Device names are listed in descending order of priority.

		M	C	P	D
a)	What is an interrupt? Describe the types of interrupts.	3	3	2	A
b)	How does Microprocessor 8086 handle the above situation in the daisy chain process? Explain the handling process with the proper figure.	7	3	2	A
2.a)	Explain how the 8259A PIC can be expanded to handle up to 64 interrupt requests without additional hardware. What is the role of the master 8259A and the eight slave 8259A controllers in this configuration?	4	3	2	A
b)	Describe the role and operation of an interrupt vector table in a Microcontrollers system. How does it manage multiple interrupts efficiently?	4	3	2	U
c)	Explain the role of cache memory in microprocessors. How do L1, L2, and L3 cache levels differ in terms of performance and usage?	2	1	1	R

Group - B

		M	C	P	D
3 a)	Write the differences between IO-mapped I/O and Memory-mapped I/O.	3	3	2	A

b)	Draw the block diagram of 8255A.	3	3	2	
c)	How can you perform Multiplication and Division operations in Microcontroller 8051? Explain the process.	4	1	1	R

OR

3 a)	What role does a Programmable Logic Device (PLD) play in decoding I/O addresses in embedded systems? How does incorporating eight additional address lines (A15–A8) influence the address decoding process, and how does the PLD produce address strobes for I/O ports?	4	3	2	A
b)	Describe the basic input and output operations within a microprocessor system. Provide a diagram to explain the data movement between the microprocessor and I/O devices.	4	3	2	N
c)	Describe the basic Direct Memory Access (DMA) cycle in microprocessor systems. How does DMA facilitate data transfer between memory and peripheral devices without involving the CPU? Explain the key steps involved in a typical DMA transfer cycle and discuss the advantages of using DMA in data-intensive applications.	2	1	1	R
4.a)	Examine the pinout diagram of the Intel 8051 microcontroller. Describe the roles of Port 0, Port 1, Port 2, and Port 3, and explain how each port enables the microcontroller to interface with external memory, input/output devices, and other peripherals.	4	2	1	A
b)	Explain how zones 1, 2, 3, and 4 are activated in an 8051 microcontroller program that displays different zone numbers on a seven-segment display. Include the logic used to determine the active zone and how this information is output to the display.	4	2	1	N
c)	Explain the key components typically integrated into a System on Chip (SoC). How do these components work together to provide a complete solution for embedded systems, and what challenges do designers face when developing SoC architectures?	2	2	1	R
5 a)	How does the handshaking or polling method synchronize the communication between a microprocessor and slower I/O devices, such as a parallel printer? Provide an example and explain the mechanisms used to slow down the microprocessor to match the speed of the printer.	3	3	2	A
b)	Develop a finite state machine(FSM) for an elevator system that operates in a multi-story building. The FSM should manage the elevator's movements, door operations, and respond to user inputs such as floor selection and door open/close commands. Explain the states and transitions involved, including how the system prioritizes requests from different floors.	4	2	1	R
c)	What is hardware-software codesign, and how does it enhance the development of embedded systems? Discuss the key benefits and challenges associated with integrating hardware and software design processes in the development lifecycle.	3	2	1	U

International Islamic University Chittagong
Center for General Education (CGED)
Semester End Examination, Spring–2024

Course Code: URED- 3503 (URED- 3101 for LLB)
Course Title: Political Thoughts and Social Behavior

Full Marks: 50 **Time: 2 Hours 30 Minutes**

(Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question):

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1	"Marriage according to Islam is a contract under the divine guidance"- explain the statement mentioning some objectives of marriage.	10	3	Remember & Create
2	Analyze the rights of women comparing them with other existing religions and communities.	10	3	Analyze & Evaluate
3	Evaluate the fundamental principles of Islamic Economic System comparing it with traditional economic systems.	10	3	Evaluate
4	Summarize the duties of children towards their parents in the light of the holy Qur'an and Sunnah.	10	3	Create
5	a) "Islamic law has provided mankind with appropriate dress code for men and women"- explain elaborately. **Or,** b) Explain any **five** of the followings: • Duty towards neighbors • Greetings • Tolerance • Honesty • Reliance on Allah • Backbiting	10	3	Create

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International Islamic University Chittagong

Morality Development Program (MDP)
(Other than Shariah Faculty)
Semester End Examination, Spring 2024
Course Title: Concepts of Moral Development II
Course Code: MDP-3505

Full Marks: 50

Time: 2 Hours & 30 Minutes

[Answer the following questions.]

1. Describe the importance of the dress code in slam with reference to the Qur'an and Hadith. What is the difference between a man and a woman's dress code in Islam? 10

2. Briefly explain the human rights in Islam. Describe the Rights of minorities and other religious groups with the reference from Qur'an and Hadith. 10

3. Describe five challenges faced by the Muslim Ummah. 10

4. What do you mean by culture? Write and discuss the features of Islamic culture. 10

Or, State any three verses of Qur'an and two Hadiths about the day of judgment. 10

5. What do you know about the Illuminati? Describe the activities of the Illuminati. 10

Or, Write the difference between Islamic culture and Western culture. Briefly describe the rights of relatives, orphans, the needy, and traveler with respect to Islam. 10